Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is: This pattern indicates a TCP SYN Flood attack (a form of Denial-of-Service attack). The server (192.0.2.1) is being bombarded with a high volume of connection requests from this IP ADDRESS : 203.0.113.0, exhausting its resources and causing the connection timeouts experienced by your employees

The logs show that: The packet logs indicate a potential TCP SYN flood attack, possibly a DDoS (Distributed Denial of Service) attack, targeting the web server on port 443. The high volume of incoming SYN requests from multiple IP addresses may be exhausting the server's resources and preventing legitimate employees from accessing the sales webpage. IP spoofing is another possible attack technique, or maybe the DDOS as we can find a lot of IP addresses but the available logs alone do not confirm it.

This event could be: This event could be a TCP SYN Flood DoS attack, potentially involving IP spoofing.. Or maybe a DDOS attack

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. SYN (Synchronize): The client sends a SYN packet to the server to request a TCP connection.
2. SYN-ACK (Synchronize-Acknowledge): The server responds with a SYN-ACK packet to acknowledge the request and indicate that it is ready to establish the connection.
3. ACK (Acknowledge): The client sends an ACK packet to confirm the server's response, and the TCP connection is established.

Explain what happens when a malicious actor sends a large number of SYN packets all at once: A malicious actor sends a large number of SYN packets to the web server without completing the TCP three-way handshake. The server allocates resources and waits for the final ACK packets, leaving many connections half-open. This can exhaust the server's resources, prevent legitimate users from connecting, and cause the website to become unavailable. This is known as a TCP SYN Flood DoS attack.. DDOS attack is another possible as we can find a lot of IP addresses.

Explain what the logs indicate and how that affects the server: The logs indicate a potential TCP SYN Flood DoS attack, with a large number of SYN packets sent to the web server on port 443 from the same and multiple IP addresses. The server responds to some requests with RST, ACK packets, suggesting that some connections are being rejected or reset. This excessive traffic can consume server resources, prevent legitimate TCP connections from being established, and cause the website to become slow or unavailable to employees and customers... IP spoofing is a possible attack technique, but the available logs alone do not confirm it.